

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Second Semester of M. Tech. (Electrical Power System) Examination****(April-May, 2018)****EE744 Advanced Power System Dynamics & Stability****Date: 01/05/2018, Tuesday****Time: 01:30 p.m. To 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1** What is the need of excitation system? Draw the functional block diagram of synchronous generator excitation system and explain the function of each component. **[07]**
- Q - 2 (A)** Draw and explain DC excitation system with amplidyne voltage regulator. **[07]**
- (B)** List out various control and protective functions of excitation system and explain each one in detail. **[07]**

OR

- Q - 2 (A)** Derive the swing equation of synchronous machine with necessary block diagram. **[07]**
- (B)** Derive the equation for synchronizing power co-efficient also state the condition for stable and unstable system. **[07]**
- Q - 3 Attempt any Two** **[14]**
- (A)** Draw and explain the steps to obtain P-V and Q-V curve for the voltage stability analysis of the power system.
- (B)** With the use of two bus network, derive the relation between receiving end power and voltage. Also, draw and explain the relationship between power, voltage and current as a function of load impedance.
- (C)** With the help of equal area criteria, explain the case of application of three phase fault at the end of two parallel lines for transient stability analysis. Derive the equation of critical clearing angle for this case.

SECTION – II

- Q - 4** Derive the block diagram of a SMIB system for small signal stability analysis using classical generator. **[07]**
- Q - 5 (A)** Derive the expression for $\tilde{E}_q$ in phasor form with d, q axes as references with required phasor diagram. **[06]**
- (B)** Draw block diagram of Thyristor excitation system with AVR and derive the complete state space model for the power system including excitation system. **[08]**

OR

Q – 5 (A) Derive the model of synchronous machine with amortisseurs neglected. **[06]**

(B) Explain the constant flux linkage model of synchronous machine with necessary diagram. **[08]**

Q – 6 Attempt any Two [14]

(A) Discuss the special case about the requirement of PSS. Give the structure of PSS and explain the function of each block of PSS.

(B) Derive the equations to analyzed transient stability of power system using point-by-point method. List out the necessary assumptions and draw the required curves for rotor speed, rotor angle and accelerating power.

(C) With zero armature resistance, Show that expressions for Δe_d and Δe_q with $p(\Delta\delta)$ and $p(\Delta\psi)$ terms included equal to those with both terms neglected.
